Data Preprocessing and Mining (Lecture - 12)



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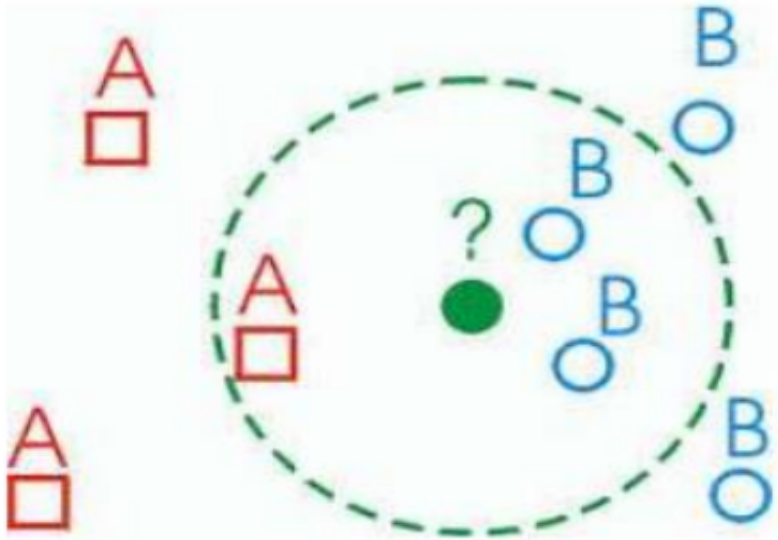
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KNN

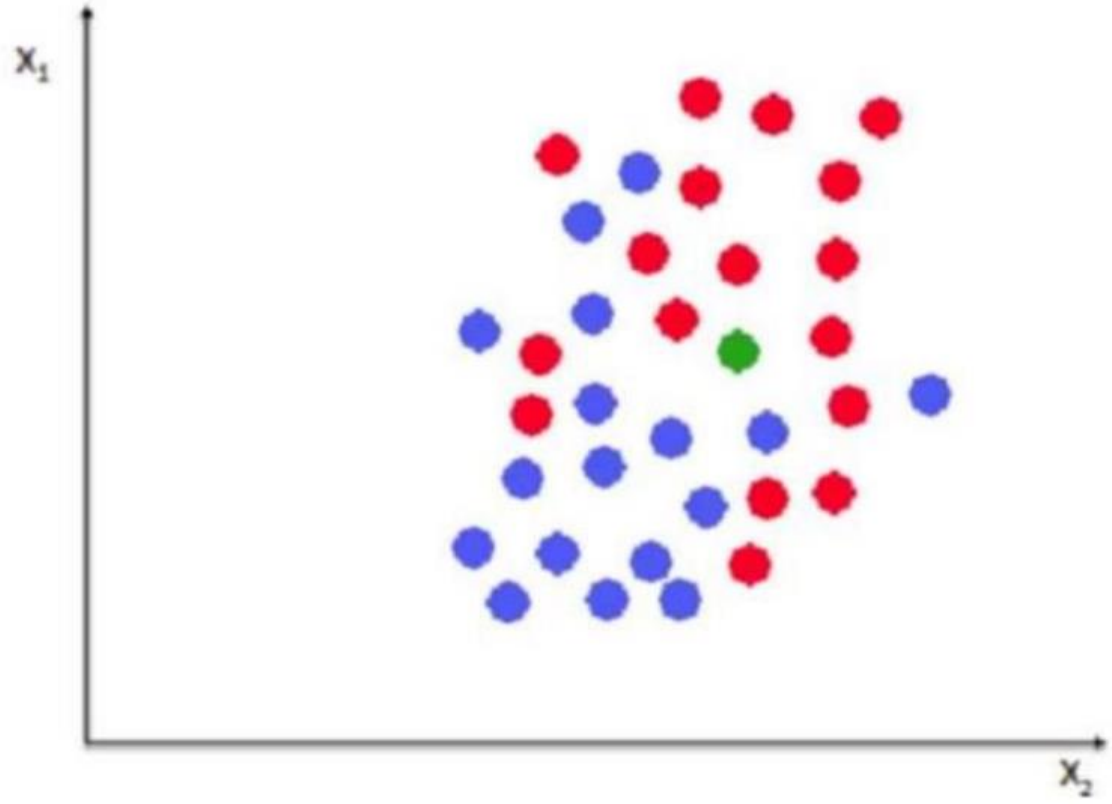
- K-Nearest Neighbors (KNN)
- Simple, but a very powerful classification algorithm
- Classifies based on a **similarity measure**
- Non-parametric
- Lazy learning
  - Does not “learn” until the test example is given
  - Whenever we have a new data to classify, we find its K-nearest neighbors from the training data

KNN: Classification Approach

- Classified by "**MAJORITY VOTES**" for its neighbor classes
  - Assigned to the most common class amongst its *K*-nearest neighbors (by measuring "distant" between data)



KNN: Example



KNN: Pseudocode

- Step 1: Determine parameter K = number of nearest neighbors
- Step 2: Calculate the distance between the query-instance and all the training examples.
- Step 3: Sort the distance and determine nearest neighbors based on the k-th minimum distance.
- Step 4:Gather the category Y of the nearest neighbors.
- Step 5: Use simple majority of the category of nearest neighbors as the prediction value of the query instance.

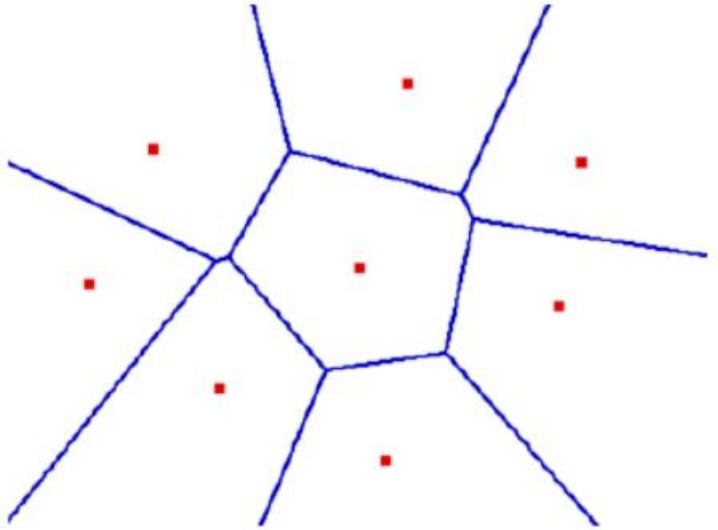
KNN: Euclidean distance matrix

Table 1. Euclidean distance matrix D listing all possible pairwise Euclidean distances between 19 samples.

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	x_{11}	x_{12}	x_{13}	x_{14}	x_{15}	x_{16}	x_{17}	x_{18}
x_2	1.5																	
x_3	1.4	1.6																
x_4	1.6	1.4	1.3															
x_5	1.7	1.4	1.5	1.5														
x_6	1.3	1.4	1.4	1.5	1.4													
x_7	1.6	1.3	1.4	1.4	1.5	1.8												
x_8	1.5	1.4	1.6	1.3	1.7	1.6	1.4											
x_9	1.4	1.3	1.4	1.5	1.2	1.4	1.3	1.5										
x_{10}	2.3	2.4	2.5	2.3	2.6	2.7	2.8	2.7	3.1									
x_{11}	2.9	2.8	2.9	3.0	2.9	3.1	2.9	3.1	3.0	1.5								
x_{12}	3.2	3.3	3.2	3.1	3.3	3.4	3.3	3.4	3.5	3.3	1.6							
x_{13}	3.3	3.4	3.2	3.2	3.3	3.4	3.2	3.3	3.5	3.6	1.4	1.7						
x_{14}	3.4	3.2	3.5	3.4	3.7	3.5	3.6	3.3	3.5	3.6	1.5	1.8	0.5					
x_{15}	4.2	4.1	4.1	4.1	4.1	4.1	4.1	4.1	4.1	4.1	1.7	1.6	0.3	0.5				
x_{16}	4.1	4.1	4.1	4.1	4.1	4.1	4.1	4.1	4.1	4.1	1.6	1.5	0.4	0.5	0.4			
x_{17}	5.9	6.2	6.2	5.8	6.1	6.0	6.1	5.9	5.8	6.0	2.3	2.3	2.5	2.3	2.4	2.5		
x_{18}	6.1	6.3	6.2	5.8	6.1	6.0	6.1	5.9	5.8	6.0	3.1	2.7	2.6	2.3	2.5	2.6	3.0	
x_{19}	6.0	6.1	6.2	5.8	6.1	6.0	6.1	5.9	5.8	6.0	3.0	2.9	2.7	2.4	2.5	2.8	3.1	0.4

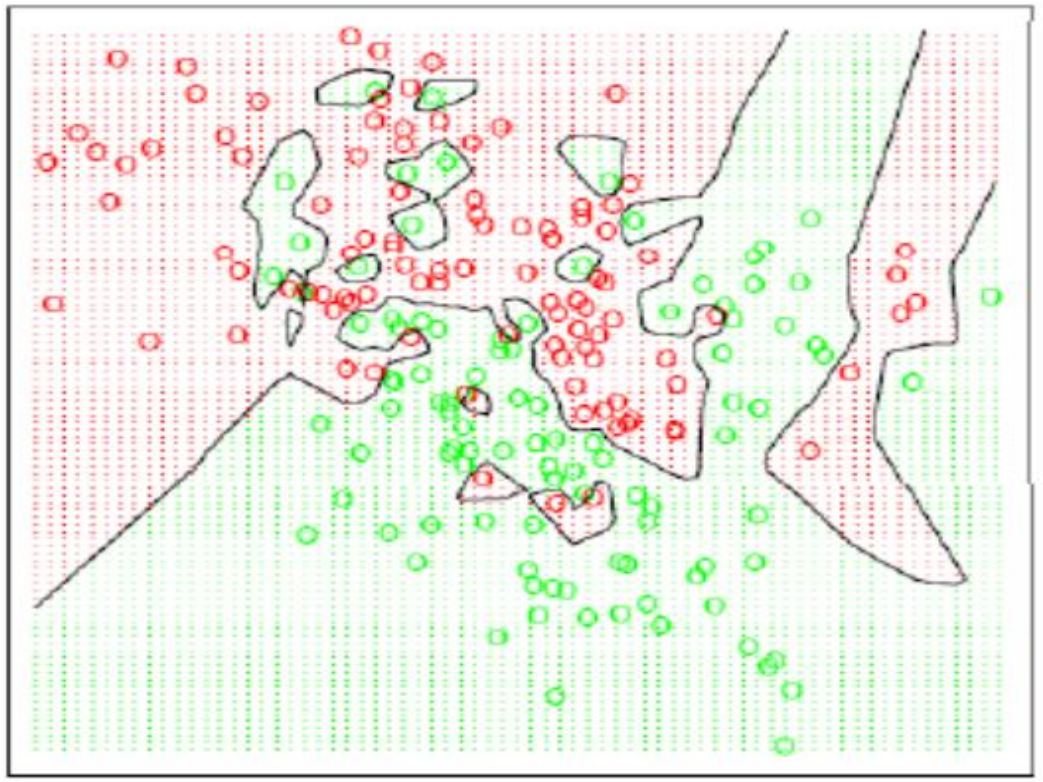
Decision Boundaries

- Voronoi diagram
  - Describes the areas that are nearest to any given point, given a set of data.
  - Each line segment is equidistant between two points of opposite class



Decision Boundaries

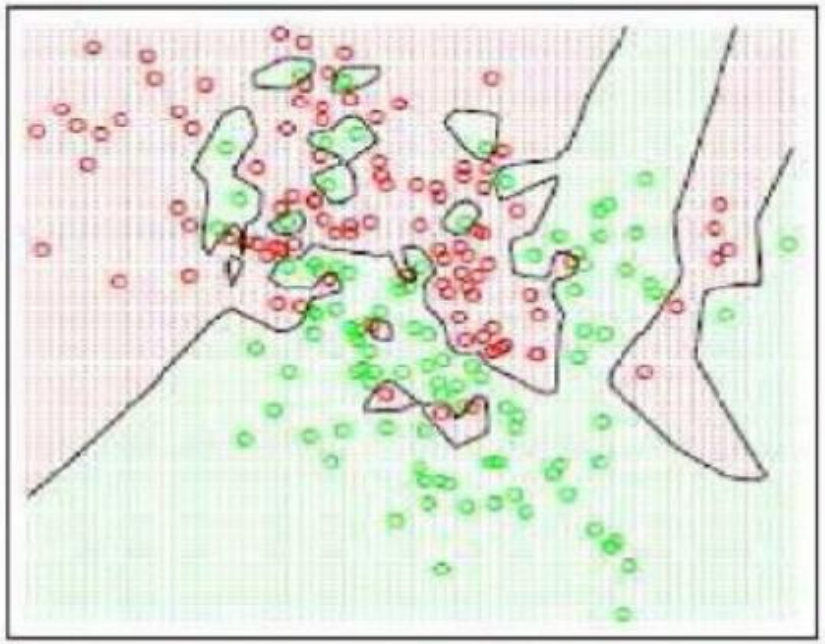
- With large number of examples and possible noise in the labels, the decision boundary can become nasty!
  - "Overfitting" problem



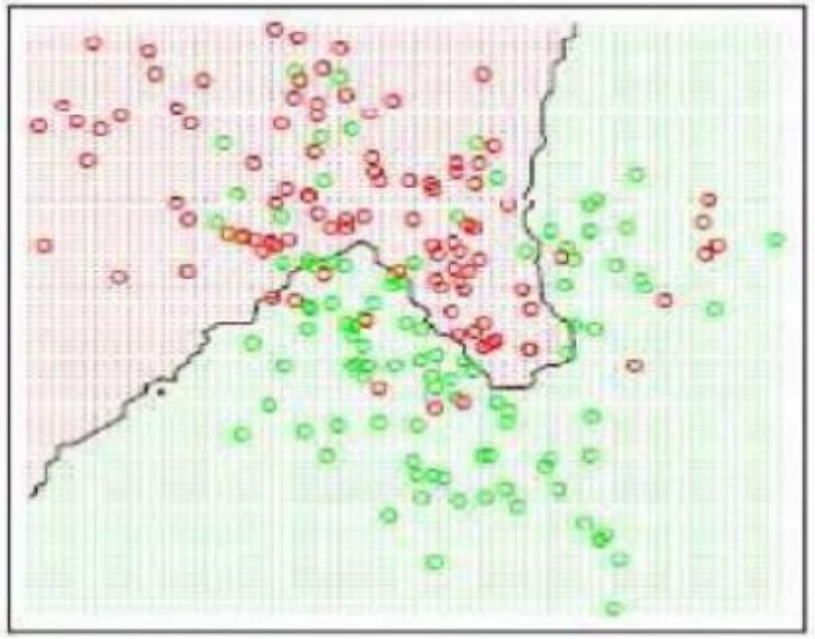
Effect of K

- Larger k produces smoother boundary effect
- When K==N, always predict the majority class

K=1



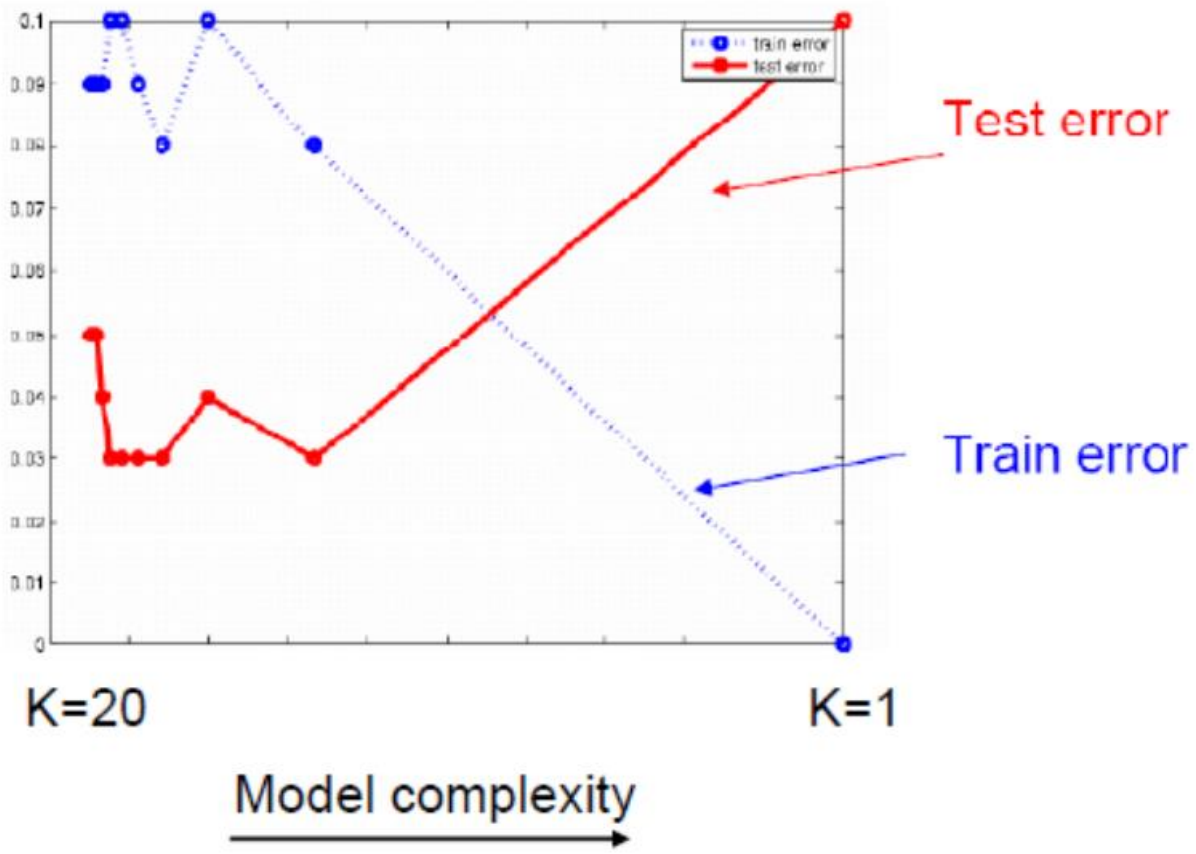
K=15



Figures from Hastie, Tibshirani and Friedman (Elements of Statistical Learning)

How to choose k?

- Empirically optimal k?



Pros and Cons

- Pros
  - Learning and implementation is extremely simple and Intuitive
  - Flexible decision boundaries
- Cons
  - Irrelevant or correlated features have high impact and must be eliminated
  - Typically difficult to handle high dimensionality
  - Computational costs: memory and classification time computation

Similarity and Dissimilarity

- Similarity
  - Numerical measure of how alike two data objects are.
  - Is higher when objects are more alike.
  - Often falls in the range [0,1]
- Dissimilarity
  - Numerical measure of how different are two data objects
  - Lower when objects are more alike
  - Minimum dissimilarity is often 0
  - Upper limit varies
- Proximity refers to a similarity or dissimilarity

Standardization

- Standardization or Z-score normalization
  - Rescale the data so that the mean is zero and the standard deviation from the mean (standard scores) is one

$$x_{norm} = \frac{x - \mu}{\sigma}$$

μ is mean, σ is a standard deviation from the mean (standard score)

Min-Max scaling

- Scale the data to a fixed range – between 0 and 1

$$x_{morm} = \frac{x - x_{min}}{x_{max} - x_{min}}$$


thank
you